

IEEE 802.3 and 802.1 groups where multiple new specifications are listed and revisions to existing specifications are done to meet automotive requirements.

Automotive Ethernet, including 100BASE-T1 and 1000BASE-T1, is not like low-voltage differential signaling (LVDS), which is point-to-point serial connectivity. Automotive Gigabit Ethernet is capable of handling multiple video streams from many cameras. Other obvious places to use 1000BASE-T1 are in connecting multiple cameras and in a variety of infotainment systems. Main driving factors are cost reduction, efficiency and reduced time-to-market.

Technology has paved the way by developing a 100Mbit per second Ethernet physical-layer standard, specifically designed for automotive connectivity applications. It can run high-bandwidth data over a single-pair unshielded twisted-pair copper wire, cutting cable and connector costs, while reducing the weight of the wiring harness inside a car.

The standard's layered approach enabled the development of new physical layer (PHY) specifications, media access protocols (MACs) and other capabilities, allowing it to keep pace with the demands of modern networks. Extensions to the original IEEE 802.1/802.3 standards now define practical solutions for providing security, deterministic performance, traffic shaping and many other application-specific functions, features and specifications. As a result, Ethernet has many capabilities that it needs to meet the unique demands of in-vehicle networking.

One-Pair Ethernet

A proprietary PHY standard, BroadR-Reach, enables longer distances of copper Ethernet connectivity at 100Mbps (up to 700 metres). This PHY uses technolo-

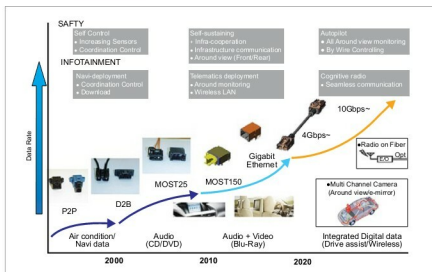


Fig. 2: Automotive market potential

gies from 1GE copper including multi-level PAM-3 signaling and better encoding (to reduce the bandwidth required on the cable) and using echo cancellers, so bi-directional data could be transmitted on a single pair. Due to the lower bandwidth (~27MHz bandwidth, 62.5MHz for 100BASE-T), this standard met automotive EMI requirements and Broadcom started marketing this to the automotive world.

As the industry realised that 100Mbps is enough for video transmission but not enough to act as a backbone in the car, it pushed for the creation of a task force for 802.3 (802.3bp) to define a standard for 1G over a single twisted pair for links of up to 15M for the automotive market. The new PHY is known as 1000BASE-T1 (where 1 stands for one pair). Based on current timelines, it is expected that prototypes will be available this year, and this will become a standard by 2017.

Power-over Ethernet

To further reduce the wiring needed in a car, Power-over Ethernet (PoE) may be used to power devices. IEEE 801.3bu (One-Pair Power-over Data Lines, or PoDL) task force is working on standardising PoE over a single pair. This is a minor

modification to the current PoE, and this standard is expected to be approved soon.

Energy-Efficient Ethernet

A car's electrical components do not all turn off when the engine is turned off. When the engine is off, battery capacity is limited. In order to minimise power consumption when the engine is off and when the engine is on, Energy-Efficient Ethernet (EEE), which turns off the network when not in use, is used. Those components that do not need to be powered on at all when the engine is off will have their network segments completely turned off. Those that need to be on will use EEE to minimise power consumption.

Gigabit Ethernet

A single-pair version of Gigabit Ethernet intended as a high-speed aggregation backbone for 1Tcpe traffic reduces cable weight. It could also serve as a direct connection between network backbone and diagnostics, ADAS cameras, infotainment video and other applications with data requirements above 100Mbps.

It meets automotive system EMC requirements, supports 15 metres over unshielded automotive cabling. It has a standard Gigabit Ethernet